

- 17 Two monoatomic ideal gases X and Y are mixed together in a sealed container. The molar mass of Y is twice that of X. At thermodynamic temperature T , the kinetic energy and root-mean-square speed of an atom of X are given by E and V respectively.

What is the kinetic energy and root-mean-square speed of an atom of Y at temperature T ?

	kinetic energy	root-mean-square speed
A	E	$0.71V$
B	E	V
C	E	$1.4V$
D	$2E$	$0.71V$

Ans: A

In general, kinetic energy of monoatomic ideal gas $E = \frac{3}{2}kT$

When temperature is T , the KE of Y is E , independent of mass.

For X at temperature T ,

$$\frac{3}{2}kT = \frac{1}{2}mV^2$$

For Y at temperature T ,

$$\frac{3}{2}k(T) = \frac{1}{2}(2m)V_y^2$$

Comparing right hand side of equation

$$\text{Thus, } V^2 = 2V_y^2$$

$$V_y = \sqrt{\frac{1}{2}V^2} = 0.71V$$

